

KA-CHOW NET: THE RADIATOR SPRINGS NETWORK

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Introduction

Ka-Chow Net is a Cisco Packet Tracer network designed for seven units of Radiator Springs: Sheriff Station (SS), Luigi Casa-Della Tires (LCT), Doc-Hudson Clinic (DC), Flo V8 Café (FVC), Ramone Body Art (RBA), Mater Tow Yard (MTY), and Wheel Well Motel (WWM).

The network implements VLSM, RIPv2, static routing, recursive and floating static routes, DHCP, DHCP relay, DNS, web services, email services, and network printers. The main goal

was to provide complete communication among all units while also providing backup routes in case of primary route failure.

1. Network Topology

The final topology contains seven routers, eight switches, two PCs and one printer per unit, and the DHCP, DNS, web, and email servers that are required.

The RIPv2 Gossip Loop consists of SS, RBA, and DC, while SS, LCT, DC, FVC, and MTY are shared by all.

The downtown network; FVC also has a direct serial connection with SS and another serial connection with WWM.

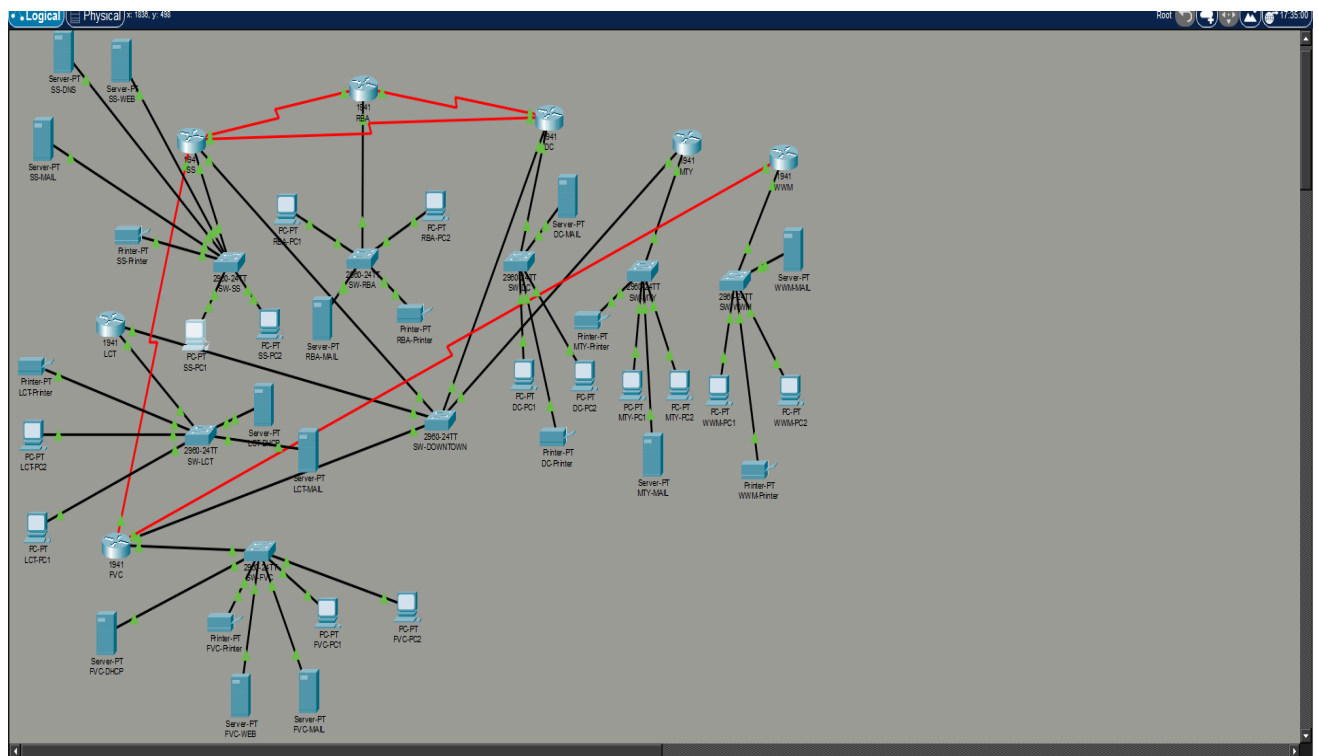


Figure 1: The complete Ka-Chow Net topology as it appears in Cisco Packet Tracer.

2. Variable Length Subnet Masking and IP Addressing

The base network was derived from the first group member's ID:

Student ID: 22201110

Last four digits:

11,10

Therefore, the base network is:

11.10.0.0/16

Address blocks were then allocated in the required order: SS, DC, FVC, LCT, RBA, MTY, WWM, and finally the inter-router segments. Larger LANs receive larger prefixes, while point-to-point links use /30 subnets.

Network	Hosts	Network Address	Prefix	Subnet Mask	Usable Host Range	Broadcast
SS	260	11.10.0.0	/23	255.255.254.0	11.10.0.1 – 11.10.1.254	11.10.1.255
DC	180	11.10.2.0	/24	255.255.255.0	11.10.2.1 – 11.10.2.254	11.10.2.255
FVC	150	11.10.3.0	/24	255.255.255.0	11.10.3.1 – 11.10.3.254	11.10.3.255
LCT	220	11.10.4.0	/24	255.255.255.0	11.10.4.1 – 11.10.4.254	11.10.4.255
RBA	90	11.10.5.0	/25	255.255.255.128	11.10.5.1 – 11.10.5.126	11.10.5.127
MTY	90	11.10.5.128	/25	255.255.255.128	11.10.5.129 – 11.10.5.254	11.10.5.255
WWM	40	11.10.6.0	/26	255.255.255.192	11.10.6.1 – 11.10.6.62	11.10.6.63
Downtown	5	11.10.6.64	/29	255.255.255.248	11.10.6.65 – 11.10.6.70	11.10.6.71
SS–RBA	2	11.10.6.72	/30	255.255.255.252	11.10.6.73 – 11.10.6.74	11.10.6.75
RBA–DC	2	11.10.6.76	/30	255.255.255.252	11.10.6.77 – 11.10.6.78	11.10.6.79
DC–SS	2	11.10.6.80	/30	255.255.255.252	11.10.6.81 – 11.10.6.82	11.10.6.83

SS-FVC	2	11.10.6.84	/30	255.255.255.252	11.10.6.85 – .86	11.10.6.87
FVC-WWM	2	11.10.6.88	/30	255.255.255.252	11.10.6.89 – .90	11.10.6.91

The allocation of the networks was carried out on the basis of VLSM in accordance with the required number of host addresses. This addressing scheme gives each unit enough host addresses while avoiding unnecessary address wastage.

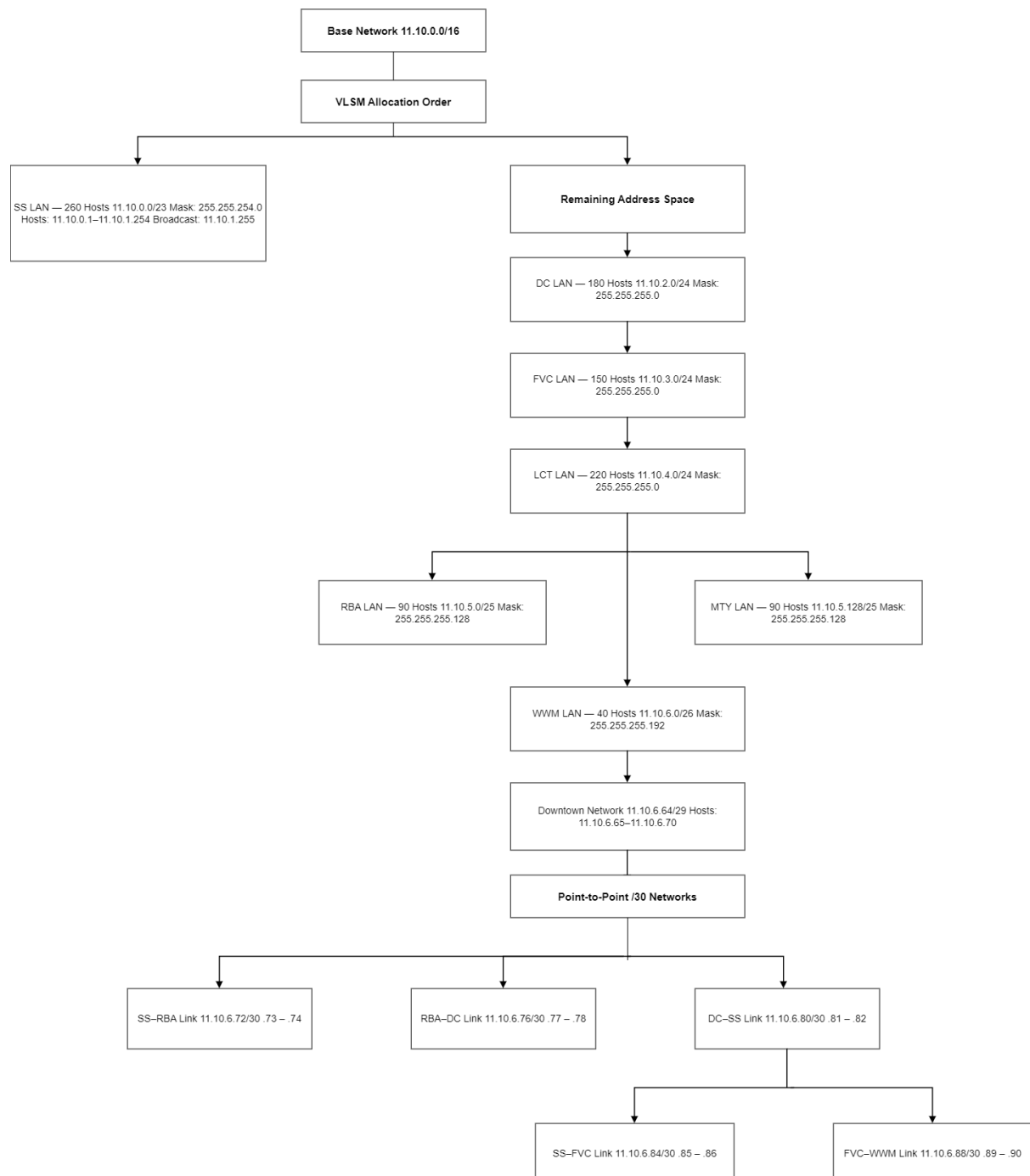


Figure 2: The complete VLSM allocation tree for the base network 11.10.0.0/16.

The VLSM tree illustrates the way in which the base network 11.10.0.0/16 was allocated from the largest. The requirements for the smaller LAN and the inter-router networks. The largest SS network received a /23 subnet, the medium LANs received /24 and /25 subnets, WWM received a /26 subnet, Downtown received a /29 subnet, and point-to-point router links were assigned /30 networks.

Unit	Device	IP Address	Subnet Mask	Default Gateway
SS	Router G0/0	11.10.0.1	255.255.254.0	N/A
SS	Router G0/1	11.10.6.65	255.255.255.248	N/A
SS	Router S0/0/0	11.10.6.73	255.255.255.252	N/A
SS	Router S0/0/1	11.10.6.81	255.255.255.252	N/A
SS	Router S0/1/0	11.10.6.85	255.255.255.252	N/A
SS	DNS Server	11.10.0.2	255.255.254.0	11.10.0.1
SS	Web Server	11.10.0.3	255.255.254.0	11.10.0.1
SS	Mail Server	11.10.0.4	255.255.254.0	11.10.0.1
SS	Printer	11.10.0.5	255.255.254.0	11.10.0.1
SS	PC1, PC2	DHCP	255.255.254.0	11.10.0.1
LCT	Router G0/0	11.10.4.1	255.255.255.0	N/A
LCT	Router G0/1	11.10.6.66	255.255.255.248	N/A
LCT	DHCP Server	11.10.4.2	255.255.255.0	11.10.4.1
LCT	Mail Server	11.10.4.3	255.255.255.0	11.10.4.1
LCT	Printer	11.10.4.4	255.255.255.0	11.10.4.1
LCT	PC1, PC2	DHCP	255.255.255.0	11.10.4.1
DC	Router G0/0	11.10.2.1	255.255.255.0	N/A
DC	Router G0/1	11.10.6.67	255.255.255.248	N/A
DC	Router S0/1/0	11.10.6.78	255.255.255.252	N/A
DC	Router S0/1/1	11.10.6.82	255.255.255.252	N/A

DC	Mail Server	11.10.2.2	255.255.255.0	11.10.2.1
DC	Printer	11.10.2.3	255.255.255.0	11.10.2.1
DC	PC1, PC2	DHCP	255.255.255.0	11.10.2.1
FVC	Router G0/0	11.10.3.1	255.255.255.0	N/A
FVC	Router G0/1	11.10.6.68	255.255.255.248	N/A
FVC	Router S0/1/0	11.10.6.86	255.255.255.252	N/A
FVC	Router S0/1/1	11.10.6.89	255.255.255.252	N/A
FVC	DHCP Server	11.10.3.2	255.255.255.0	11.10.3.1
FVC	Web Server	11.10.3.3	255.255.255.0	11.10.3.1
FVC	Mail Server	11.10.3.4	255.255.255.0	11.10.3.1
FVC	Printer	11.10.3.5	255.255.255.0	11.10.3.1
FVC	PC1, PC2	DHCP	255.255.255.0	11.10.3.1
RBA	Router G0/0	11.10.5.1	255.255.255.128	N/A
RBA	Router S0/1/0	11.10.6.74	255.255.255.252	N/A
RBA	Router S0/1/1	11.10.6.77	255.255.255.252	N/A
RBA	Mail Server	11.10.5.2	255.255.255.128	11.10.5.1
RBA	Printer	11.10.5.3	255.255.255.128	11.10.5.1
RBA	PC1, PC2	DHCP	255.255.255.128	11.10.5.1
MTY	Router G0/0	11.10.5.129	255.255.255.128	N/A
MTY	Router G0/1	11.10.6.69	255.255.255.248	N/A
MTY	Mail Server	11.10.5.130	255.255.255.128	11.10.5.129
MTY	Printer	11.10.5.131	255.255.255.128	11.10.5.129
MTY	PC1, PC2	DHCP	255.255.255.128	11.10.5.129
WWM	Router G0/0	11.10.6.1	255.255.255.192	N/A
WWM	Router S0/1/0	11.10.6.90	255.255.255.252	N/A
WWM	Mail Server	11.10.6.2	255.255.255.19	11.10.6.1

			2	
WWM	Printer	11.10.6.3	255.255.255.19 2	11.10.6.1
WWM	PC1, PC2	DHCP	255.255.255.19 2	11.10.6.1

Full IP addressing table for Ka-Chow Net. PCs obtain their addresses dynamically through DHCP, while routers, servers, and printers use static addresses. Router interfaces do not require a default gateway.

3. Routing Configuration

RIPv2

RIPv2 was set up on SS, RBA and DC, with version 2 being used and auto-summary disabled. LAN interfaces were kept passive so that unnecessary RIP advertisements were not sent toward end devices.

SS also redistributed its required static routes into RIP so that RBA and DC could learn routes toward networks outside the Gossip Loop.

```

SS#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 20 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip, static
Default version control: send version 2, receive 2
  Interface          Send Recv Triggered RIP Key-chain
  Serial0/0/0         22
  Serial0/0/1         22
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
  11.0.0.0
Passive Interface(s):
  GigabitEthernet0/0
  GigabitEthernet0/1
  Serial0/1/0
Routing Information Sources:
  Gateway         Distance      Last Update
  11.10.6.82       120           00:00:27
  11.10.6.74       120           00:00:05
--More--

```

Figure 3: RIPv2 configuration on Sheriff Station showing version 2 and passive interfaces.

The output confirms that SS is running RIPv2, automatic summarization is disabled, and the required interfaces are passive.

```

SS>enable
SS#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    11.0.0.0/8 is variably subnetted, 18 subnets, 7 masks
C       11.10.0.0/23 is directly connected, GigabitEthernet0/0
L       11.10.0.1/32 is directly connected, GigabitEthernet0/0
R       11.10.2.0/24 [120/1] via 11.10.6.82, 00:00:07, Serial0/0/1
S       11.10.3.0/24 [1/0] via 11.10.6.86
S       11.10.4.0/24 [1/0] via 11.10.6.66
R       11.10.5.0/25 [120/1] via 11.10.6.74, 00:00:10, Serial0/0/0
S       11.10.5.128/25 [1/0] via 11.10.6.69
S       11.10.6.0/26 [1/0] via 11.10.6.86
C       11.10.6.64/29 is directly connected, GigabitEthernet0/1
L       11.10.6.65/32 is directly connected, GigabitEthernet0/1
C       11.10.6.72/30 is directly connected, Serial0/0/0
--More-- |

```

Figure 4: SS routing table showing connected, RIP-learned, and static routes.

The routing table confirms that SS has learned RIPv2 networks while also keeping the required connected and static routes.

4. DHCP Configuration

All personal computers get their IPv4 configuration automatically through the use of DHCP.

The Sheriff Station functions as the DHCP server for the SS, RBA, and DC LANs. RBA and DC use an ip helper address to forward DHCP requests to SS.

LCT has a DHCP server of its own and at the same time supplies addresses to MTY via DHCP relay.

FVC has a separate DHCP server and provides addresses to WWM through another DHCP relay.

The central DNS server address distributed through DHCP is:

11.10.0.2

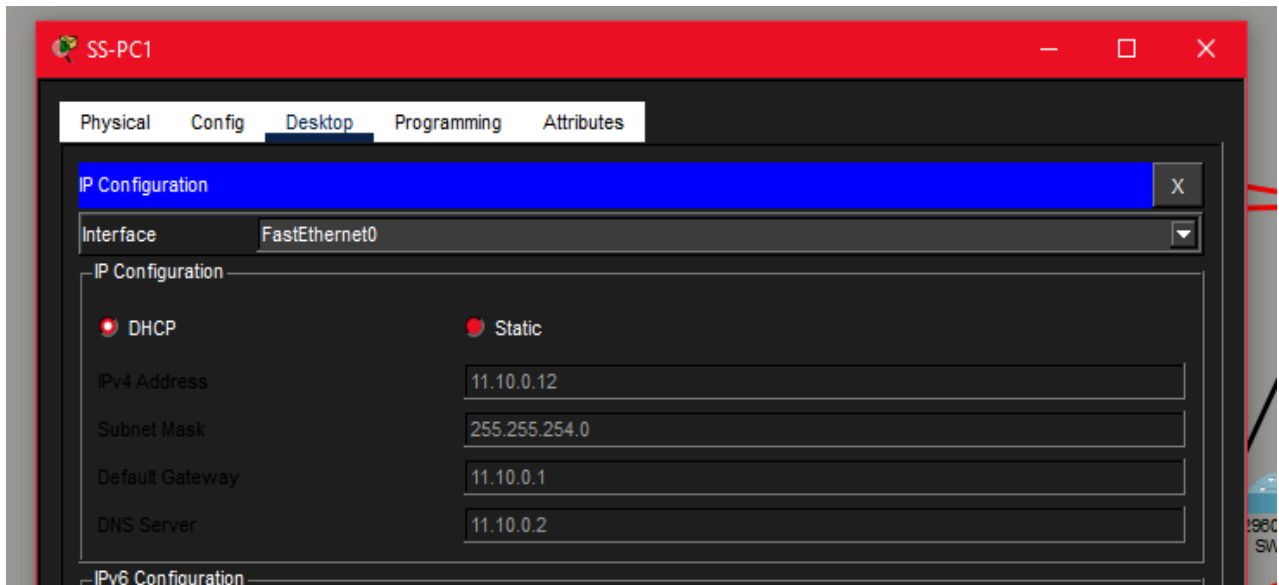


Figure 5: SS-PC1 successfully receiving its network configuration through DHCP.

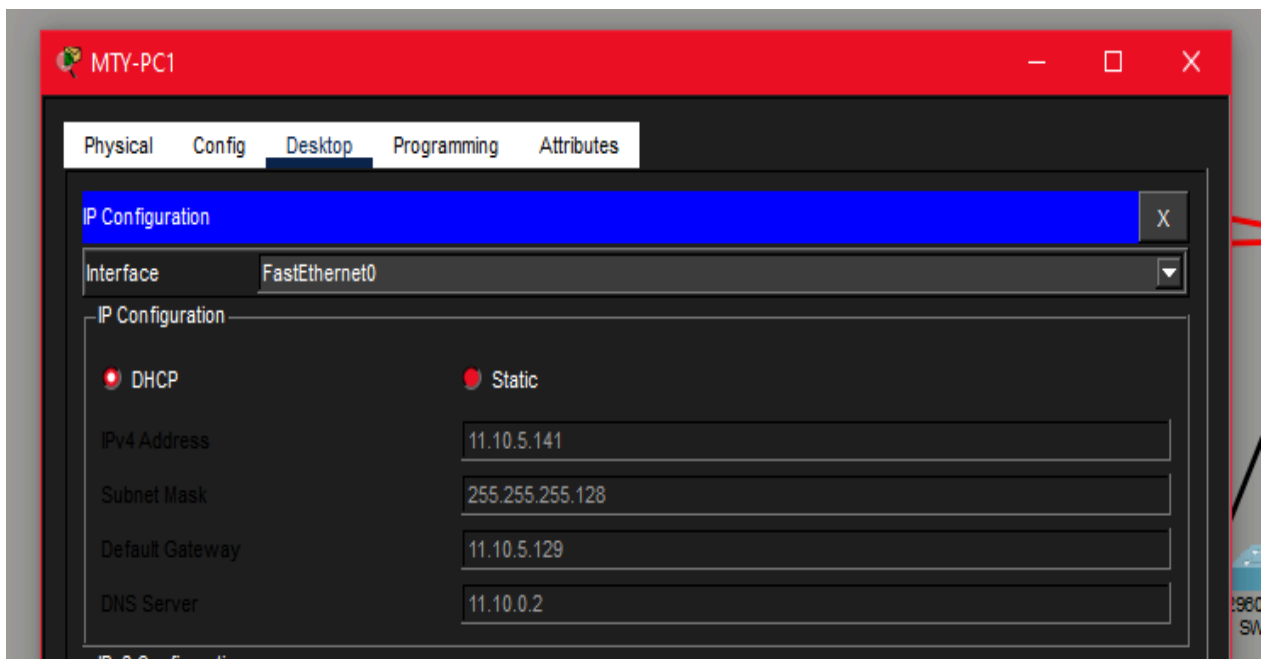


Figure 6: MTY-PC1 receiving DHCP configuration from the LCT DHCP server through relay.



Figure 7: WWM-PC1 receiving DHCP configuration from the FVC DHCP server through relay.

Sheriff Station offers DHCP service for the SS, RBA, and DC LANs. RBA and DC use ip helper-address to forward DHCP requests to the SS. The LCT makes use of its own DHCP server at 11.10.4.2 for the LCT LAN and also serves MTY through DHCP relay. FVC uses its dedicated DHCP server at 11.10.3.2 and serves WWM through relay. The screenshots confirm correct assignment of IPv4 address, subnet mask, default gateway, and the central DNS server 11.10.0.2

5. DNS, Web and Email Services

A single central DNS server was configured at Sheriff Station with IP address:

11.10.0.2

DNS records were created for the Sheriff and Flo websites as well as the email servers of all units.

No	Name	Type	Detail
0	flo.rs	A Record	11.10.3.4
1	mail.doc.rs	A Record	11.10.2.2
2	mail.flo.rs	A Record	11.10.3.4
3	mail.lugi.rs	A Record	11.10.4.3
4	mail.mater.rs	A Record	11.10.5.130
5	mail.ramone.rs	A Record	11.10.5.2
6	mail.sheriff.rs	A Record	11.10.0.4
7	mail.wheelwest.rs	A Record	11.10.6.2
8	sheriff.rs	A Record	11.10.0.4
9	www.flo.radiatorsprings.nyc	A Record	11.10.3.3
10	www.flo.v8.radiatorsprings.nyc	A Record	11.10.3.3
11	www.sheriff.radiatorsprings.nyc	A Record	11.10.0.3

Figure 8: Central DNS records configured on the Sheriff Station DNS server.

At the Sheriff Station a single central DNS server was set up with the IP address 11.10.0.2. DNS records were created for the Sheriff and Flo web servers and for the email servers of all.

There are seven units; the Sheriff web server has the address 11.10.0.3 and the Flo web server uses 11.10.3.3.

HTTP services were also enabled on both of them, and SMTP and POP3 services were enabled on each one.

A local email server, enabling two-way cross-domain email communication between sheriff.rs and flo.rs, was successfully tested.



Figure 9: Successful access to the Flo V8 Café web server using its DNS name.

6. Cross-Domain Email Communication

Each unit had its own local email server, and the main cross-domain email
The requirement was tested between the Sheriff Station and the Flo V8 Café.

Sheriff Station uses the domain:

sheriff.rs

Flo V8 Café uses:

flo.rs

A message from sheriff1@sheriff.rs was successfully received by flo1@flo.rs.

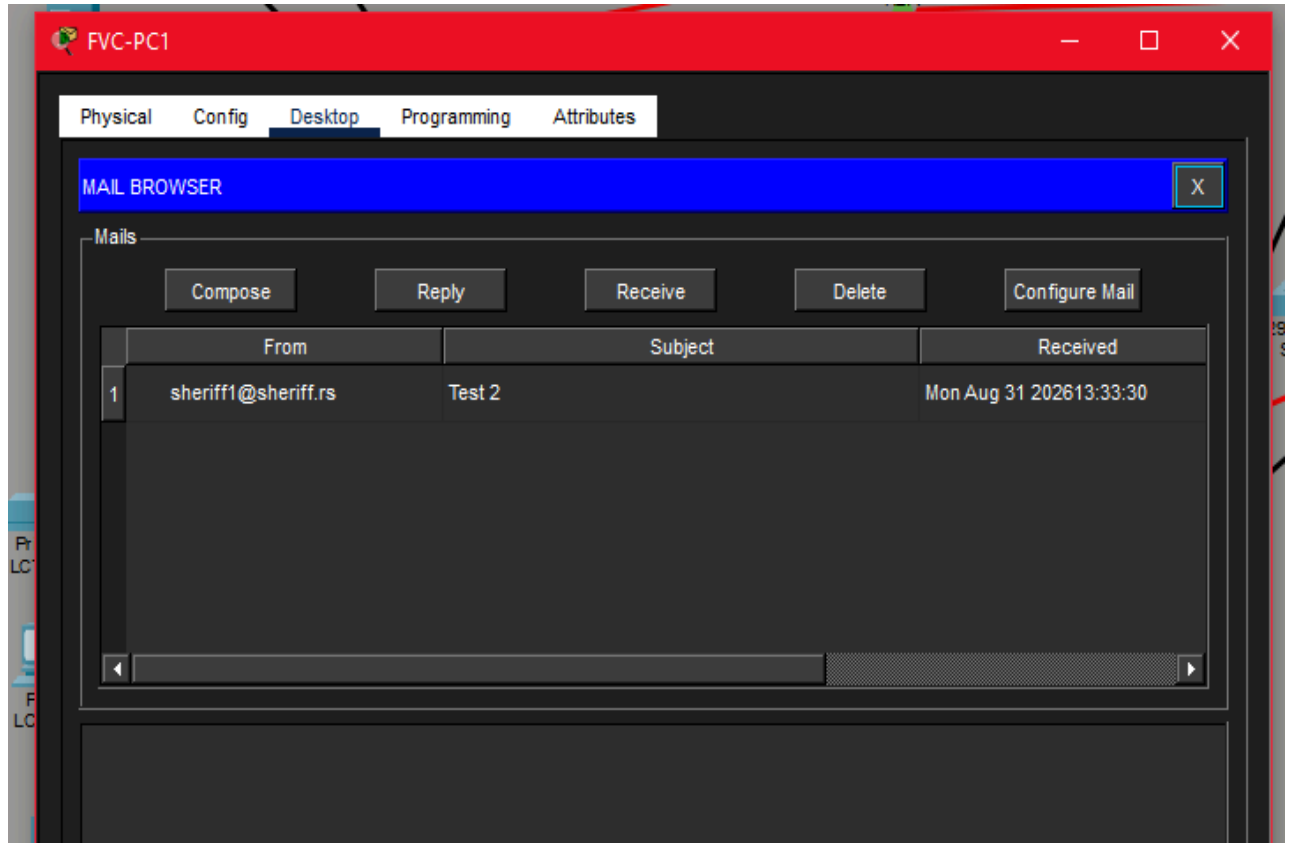


Figure 10: The Email sent from The Sheriff Station successfully received at Flo V8 Café.

The message sent from **flo1@flo.rs** was successfully received by **sheriff1@sheriff.rs**, thus establishing a two-way cross-domain email communication.

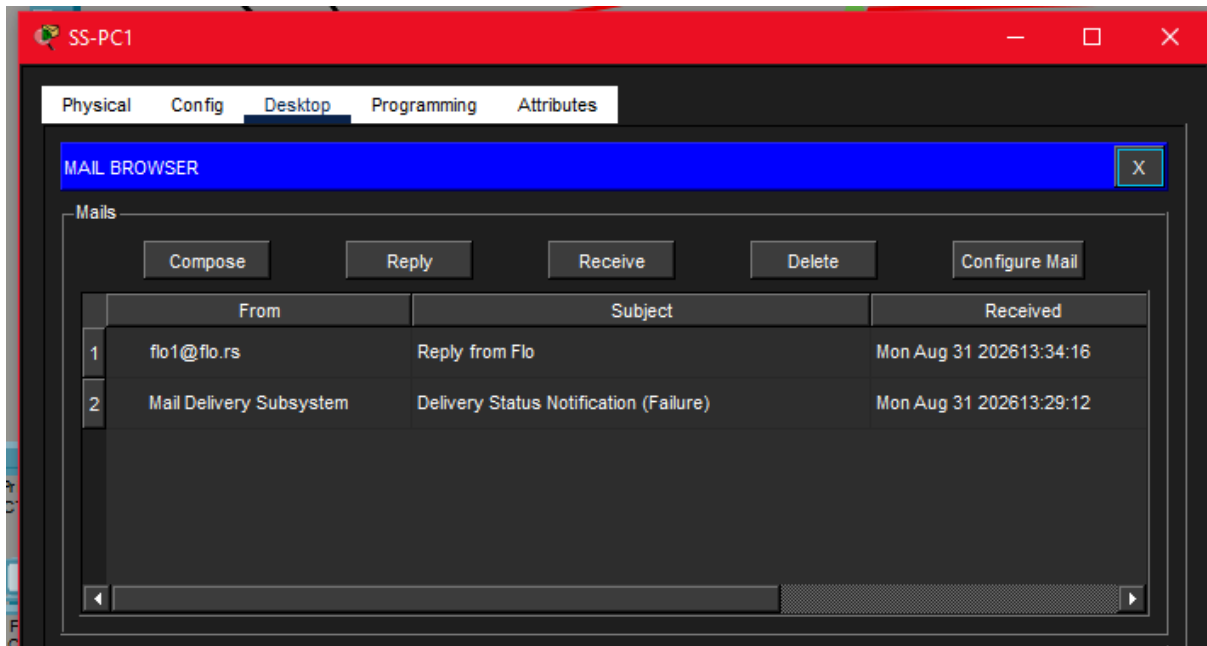


Figure 11: Reply from Flo V8 Café successfully received at Sheriff Station.

The old “Mail Delivery Subsystem” failure shown in the SS screenshot was generated during a previous attempt at the configuration attempt. The successful message above it proves that the final configuration is working correctly.

7. FVC Recursive Backup Route

FVC normally reaches the Sheriff Station network through the direct SS–FVC serial connection.

Under normal conditions, the primary static route has an administrative distance of 1 and uses Serial0/1/0.

```
FVC>show ip route 11.10.0.0
Routing entry for 11.10.0.0/23
Known via "static", distance 1, metric 0 (connected)
Routing Descriptor Blocks:
* directly connected, via Serial0/1/0
  Route metric is 0, traffic share count is 1
FVC>
```

Figure 12: FVC using the primary route toward Sheriff Station through Serial0/1/0.

A backup recursive static route was also configured through LCT using next-hop address:

11.10.6.66

The backup route uses administrative distance **5**, which keeps it inactive while the primary route is available.

When the direct FVC–SS serial link was shut down, the primary route disappeared and the backup route through LCT automatically became active.

```
FVC>enable
FVC#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
FVC(config)#interface serial0/1/0
FVC(config-if)#shutdown

FVC(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to down
end
FVC#
%SYS-5-CONFIG_I: Configured from console by console
show ip route 11.10.0.0
Routing entry for 11.10.0.0/23
Known via "static", distance 5, metric 0
  Routing Descriptor Blocks:
    * 11.10.6.66
      Route metric is 0, traffic share count is 1
```

Figure 13: FVC recursive backup route activated with AD 5 through 11.10.6.66.

Connectivity was then tested again.

```
FVC#ping 11.10.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 11.10.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

FVC#traceroute 11.10.0.1
Type escape sequence to abort.
Tracing the route to 11.10.0.1

 0  11.10.6.66      0 msec    0 msec    1 msec
 1  11.10.6.65      1 msec    0 msec    1 msec
FVC#
```

Figure 14: Successful backup-path ping and traceroute from FVC toward SS.

The traceroute shows:

FVC to 11.10.6.66 (LCT) to 11.10.6.65 (SS)

This proves that communication continued through the backup path after the direct link failed.

8. LCT Floating Static Route

LCT normally reaches the SS LAN through next hop:

11.10.6.65

The primary route has administrative distance **1**.

```
LCT>enable
LCT#show ip route 11.10.0.0
Routing entry for 11.10.0.0/23
Known via "static", distance 1, metric 0
  Routing Descriptor Blocks:
    * 11.10.6.65
      Route metric is 0, traffic share count is 1
LCT#
```

Figure 15: LCT primary static route toward Sheriff Station with AD 1.

A floating static backup route was configured through DC using next hop:

11.10.6.67

with administrative distance:

5

For failover testing, the primary route was temporarily removed. The backup route immediately became active and a successful ping to the SS gateway confirmed connectivity.


```

LCT#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
LCT(config)#no ip route 11.10.0.0 255.255.254.0 11.10.6.65
LCT(config)#end
LCT#
%SYS-5-CONFIG_I: Configured from console by console
show ip route 11.10.0.0
Routing entry for 11.10.0.0/23
Known via "static", distance 5, metric 0
  Routing Descriptor Blocks:
    * 11.10.6.67
      Route metric is 0, traffic share count is 1

LCT#ping 11.10.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 11.10.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

```

Figure 16: LCT floating static route through DC with AD 5 and successful connectivity test.

The first main route was restored following the test.

9. Final Testing

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 11.10.2.2

Pinging 11.10.2.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 11.10.2.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 11.10.3.3

Pinging 11.10.3.3 with 32 bytes of data:

Request timed out.
Reply from 11.10.3.3: bytes=32 time=2ms TTL=126
Reply from 11.10.3.3: bytes=32 time=1ms TTL=126
Reply from 11.10.3.3: bytes=32 time=1ms TTL=126

Ping statistics for 11.10.3.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>ping 11.10.4.3

Pinging 11.10.4.3 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 11.10.4.3: bytes=32 time<1ms TTL=126
Reply from 11.10.4.3: bytes=32 time=1ms TTL=126

Ping statistics for 11.10.4.3:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 11.10.5.2

Pinging 11.10.5.2 with 32 bytes of data:

Request timed out.
Reply from 11.10.5.2: bytes=32 time=11ms TTL=126
Reply from 11.10.5.2: bytes=32 time=1ms TTL=126
Reply from 11.10.5.2: bytes=32 time=13ms TTL=126

Ping statistics for 11.10.5.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 13ms, Average = 8ms

```

Figure 17: Successful ping tests from SS-PC1 to DC, FVC, and LCT.

```
C:\>ping 11.10.5.2

Pinging 11.10.5.2 with 32 bytes of data:

Request timed out.
Reply from 11.10.5.2: bytes=32 time=11ms TTL=126
Reply from 11.10.5.2: bytes=32 time=1ms TTL=126
Reply from 11.10.5.2: bytes=32 time=13ms TTL=126

Ping statistics for 11.10.5.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 13ms, Average = 8ms

C:\>ping 11.10.5.130

Pinging 11.10.5.130 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 11.10.5.130: bytes=32 time=11ms TTL=126
Reply from 11.10.5.130: bytes=32 time=1ms TTL=126

Ping statistics for 11.10.5.130:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 11ms, Average = 6ms

C:\>ping 11.10.6.2

Pinging 11.10.6.2 with 32 bytes of data:

Request timed out.
Reply from 11.10.6.2: bytes=32 time=21ms TTL=125
Reply from 11.10.6.2: bytes=32 time=11ms TTL=125
Reply from 11.10.6.2: bytes=32 time=2ms TTL=125

Ping statistics for 11.10.6.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 21ms, Average = 11ms

C:\>
```

Figure 18: Successful ping tests from SS-PC1 to RBA, MTY, and WWM.

The successful replies confirm that traffic can travel between Sheriff Station and all major remote LANs through the configured dynamic and static routing paths.

11. Conclusion

The Ka-Chow Net network was successfully implemented in Cisco Packet Tracer using the required networking concepts. VLSM was used in order to allocate the 11.10.0.0/16 address space, in an efficient manner among the seven units and inter-router networks.

RIPv2 successfully provided dynamic routing within the Gossip Loop, while static, default, exit-interface, recursive, and floating static routes were used for the remaining connectivity requirements.

DHCP and DHCP relay successfully provided automatic network configuration to all PCs. Central DNS, web services, local email servers, and two-way Sheriff–Flo email communication were also successfully implemented.

Finally, both the FVC recursive route and LCT floating static route were tested successfully, demonstrating network redundancy. End-to-end tests confirmed that the different units, servers, and network printers could communicate successfully.

12. Appendix A — Equivalent Router Configuration Commands

The following are the equivalent final configuration commands used on the seven routers. Only configuration-relevant commands are shown; default Cisco IOS lines are omitted.

A.1 Sheriff Station - SS

```
hostname SS
no ip domain-lookup

interface GigabitEthernet0/0
 ip address 11.10.0.1 255.255.254.0
 no shutdown

interface GigabitEthernet0/1
 ip address 11.10.6.65 255.255.255.248
 no shutdown

interface Serial0/0/0
 ip address 11.10.6.73 255.255.255.252
 clock rate 64000
 no shutdown

interface Serial0/0/1
 ip address 11.10.6.81 255.255.255.252
 clock rate 64000
 no shutdown

interface Serial0/1/0
 ip address 11.10.6.85 255.255.255.252
 clock rate 64000
 no shutdown

ip dhcp excluded-address 11.10.0.1 11.10.0.10
ip dhcp excluded-address 11.10.2.1 11.10.2.10
ip dhcp excluded-address 11.10.5.1 11.10.5.10

ip dhcp pool SS-LAN
 network 11.10.0.0 255.255.254.0
 default-router 11.10.0.1
 dns-server 11.10.0.2

ip dhcp pool DC-LAN
 network 11.10.2.0 255.255.255.0
 default-router 11.10.2.1
 dns-server 11.10.0.2

ip dhcp pool RBA-LAN
 network 11.10.5.0 255.255.255.128
 default-router 11.10.5.1
 dns-server 11.10.0.2

ip route 11.10.3.0 255.255.255.0 11.10.6.86
ip route 11.10.4.0 255.255.255.0 11.10.6.66
ip route 11.10.5.128 255.255.255.128 11.10.6.69
ip route 11.10.6.0 255.255.255.192 11.10.6.86
ip route 11.10.6.88 255.255.255.252 11.10.6.86

router rip
 version 2
 no auto-summary
 network 11.0.0.0
 passive-interface GigabitEthernet0/0
```

```
passive-interface GigabitEthernet0/1
passive-interface Serial0/1/0
redistribute static metric 1
```

A.2 Ramone Body Art - RBA

```
hostname RBA
no ip domain-lookup

interface GigabitEthernet0/0
 ip address 11.10.5.1 255.255.255.128
 ip helper-address 11.10.6.73
 no shutdown

interface Serial0/1/0
 ip address 11.10.6.74 255.255.255.252
 no shutdown

interface Serial0/1/1
 ip address 11.10.6.77 255.255.255.252
 clock rate 64000
 no shutdown

router rip
 version 2
 no auto-summary
 network 11.0.0.0
 passive-interface GigabitEthernet0/0
```

A.3 Doc-Hudson Clinic - DC

```
hostname DC
no ip domain-lookup

interface GigabitEthernet0/0
 ip address 11.10.2.1 255.255.255.0
 ip helper-address 11.10.6.81
 no shutdown

interface GigabitEthernet0/1
 ip address 11.10.6.67 255.255.255.248
 no shutdown

interface Serial0/1/0
 ip address 11.10.6.78 255.255.255.252
 no shutdown

interface Serial0/1/1
 ip address 11.10.6.82 255.255.255.252
 no shutdown

router rip
 version 2
 no auto-summary
 network 11.0.0.0
 passive-interface GigabitEthernet0/0
 passive-interface GigabitEthernet0/1
```

A.4 Flo V8 Cafe - FVC

```
hostname FVC
no ip domain-lookup

interface GigabitEthernet0/0
 ip address 11.10.3.1 255.255.255.0
 no shutdown

interface GigabitEthernet0/1
 ip address 11.10.6.68 255.255.255.248
 no shutdown
```

```

interface Serial0/1/0
 ip address 11.10.6.86 255.255.255.252
 no shutdown

interface Serial0/1/1
 ip address 11.10.6.89 255.255.255.252
 clock rate 64000
 no shutdown

ip route 11.10.0.0 255.255.254.0 Serial0/1/0
ip route 11.10.0.0 255.255.254.0 11.10.6.66 5
ip route 11.10.2.0 255.255.255.0 GigabitEthernet0/1
ip route 11.10.4.0 255.255.255.0 GigabitEthernet0/1
ip route 11.10.5.0 255.255.255.128 Serial0/1/0
ip route 11.10.5.128 255.255.255.128 GigabitEthernet0/1
ip route 11.10.6.0 255.255.255.192 Serial0/1/1

```

A.5 Luigi Casa-Della Tires — LCT

```

hostname LCT
no ip domain-lookup

interface GigabitEthernet0/0
 ip address 11.10.4.1 255.255.255.0
 no shutdown

interface GigabitEthernet0/1
 ip address 11.10.6.66 255.255.255.248
 no shutdown

ip route 11.10.0.0 255.255.254.0 11.10.6.65
ip route 11.10.0.0 255.255.254.0 11.10.6.67 5
ip route 11.10.2.0 255.255.255.0 11.10.6.67
ip route 11.10.3.0 255.255.255.0 11.10.6.68
ip route 11.10.5.0 255.255.255.128 11.10.6.65
ip route 11.10.5.128 255.255.255.128 11.10.6.69
ip route 11.10.6.0 255.255.255.192 11.10.6.68
ip route 11.10.6.72 255.255.255.252 11.10.6.65
ip route 11.10.6.76 255.255.255.252 11.10.6.67
ip route 11.10.6.80 255.255.255.252 11.10.6.67
ip route 11.10.6.84 255.255.255.252 11.10.6.68
ip route 11.10.6.88 255.255.255.252 11.10.6.68

```

A.6 Mater Tow Yard — MTY

```

hostname MTY
no ip domain-lookup

interface GigabitEthernet0/0
 ip address 11.10.5.129 255.255.255.128
 ip helper-address 11.10.4.2
 no shutdown

interface GigabitEthernet0/1
 ip address 11.10.6.69 255.255.255.248
 no shutdown

ip route 11.10.0.0 255.255.254.0 11.10.6.65
ip route 11.10.2.0 255.255.255.0 11.10.6.67
ip route 11.10.3.0 255.255.255.0 11.10.6.68
ip route 11.10.4.0 255.255.255.0 11.10.6.66
ip route 11.10.5.0 255.255.255.128 11.10.6.65
ip route 11.10.6.0 255.255.255.192 11.10.6.68
ip route 11.10.6.72 255.255.255.252 11.10.6.65
ip route 11.10.6.76 255.255.255.252 11.10.6.67
ip route 11.10.6.80 255.255.255.252 11.10.6.67
ip route 11.10.6.84 255.255.255.252 11.10.6.68
ip route 11.10.6.88 255.255.255.252 11.10.6.68

```

A.7 Wheel Well Motel — WWM

```
hostname WWM
no ip domain-lookup

interface GigabitEthernet0/0
 ip address 11.10.6.1 255.255.255.192
 ip helper-address 11.10.3.2
 no shutdown

interface Serial0/1/0
 ip address 11.10.6.90 255.255.255.252
 no shutdown

ip route 0.0.0.0 0.0.0.0 11.10.6.89
```